PerfectFooties — Product QA & Testing Document

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Project: PerfectFooties E-Commerce Platform
Stack: React + Vite · Firebase (Firestore, Auth, Storage, Functions) · MUI · Paystack · Mailtrap

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1. Business Overview

PerfectFooties is a Nigerian luxury leather-goods e-commerce brand that handcrafts bespoke shoes, bags, belts, wallets, and accessories. The platform serves three customer segments:

Segment	Behaviour
Retail buyers	Browse catalog, add to cart, pay via Paystack, receive items
Custom order clients	Request bespoke pieces specifying type, colour, size, and reference photos; finalise via WhatsApp

> **Note:** PerfectFooties is a product-only platform. There are no service offerings, appointments, or booking flows.

Core business rules that testers must understand:

- Production time is **10–14 days** before shipping.
- Shipping is **2–5 days** (local) or **5–10 days** (international) after production.
- Custom orders are initiated in the app and finalised over WhatsApp — no payment is taken in-app for custom orders.
- Loyalty points are earned at **15 pts per retail/product order** (status `received`).
- Every **50 pts = ₦500** discount on a future order.
- Referrers receive **50 pts** when their referral code is used; the referred customer receives a **₦500 discount**.
- Gift cards are admin-activated; one card per order at checkout.

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2. Test Environment Setup

Prerequisites

Requirement	Detail
Browser	Chrome (latest), Firefox, Safari (for iOS sim)
Devices	Desktop 1440px, tablet 768px, mobile 375px
Test payment	Use Paystack **test mode** cards (see Section 13)
Email inspection	Mailtrap inbox — ask admin for credentials

Requirement	Detail
Firebase Console	Access to `perfect-footies` project for Firestore/Storage inspection
Test email accounts	Two separate Gmail accounts (one admin, one regular user)

Environment URL

```
Production: https://perfectfooties.com
```

> All testing should begin on the live URL unless a staging environment is provisioned.

Firebase Project

```
Project ID: perfect-footies
Region: us-central1
```

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3. Test Accounts & Credentials

Role	Account	Notes
Admin user	`chizobaezeh338@gmail.com` or `perfectfooties@gmail.com`	`praiseolusegun19@gmail.com`
Regular user A	Any Google account	For placing orders, viewing account
Regular user B	Second Google account	For testing referral code flow between two users
Mailtrap inbox	Provided by dev	Inspect all transactional emails

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4. Feature Map

The following features exist and must be tested end-to-end:

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Public Features
├── Navigation (Navbar + Mobile Drawer)
├── Homepage (Hero, Collections, FAQ, Contact)
├── Shop / Collection Browsing
├── Item Detail (size, colour, add to cart)
├── Cart (add, remove, update quantity)
├── Checkout (shipping, discounts, Paystack payment)
├── Thank You Page (order summary, progress tracker)
├── Custom Order (form, photo upload, WhatsApp navigation)
```

```
  ├── Account (profile, order history with status, wishlist)
  ├── Order Detail Page (tracking, receipt download)
  ├── Gift Card Redemption (at checkout)
  ├── Loyalty Points Redemption (at checkout)
  ├── Referral Code (URL capture + checkout entry)
  ├── Newsletter Subscribe (footer)
  ├── Testimonials (submit review after delivery)
  ├── Blog (read posts)
  ├── Gallery
  ├── Wishlist (save/remove products)
  └── Dark / Light Mode

Admin Features (/admin — admin accounts only)
  ├── Dashboard (metrics, charts, quick actions)
  ├── Orders (view all, change status, notes, email)
  ├── Production Queue (in-production tracking, mark shipped)
  ├── Customers (loyalty tiers, points, order count)
  ├── Collections (products CRUD)
  ├── Newsletter (compose, send, campaign history)
  ├── Gift Cards (create, activate, track)
  ├── Loyalty & Referrals (leaderboard, manual adjustments)
  ├── Blog Posts (CRUD)
  ├── Gallery (CRUD)
  ├── Announcements (CRUD, activate/deactivate)
  └── Cancellation Requests

Cloud Functions
  ├── Paystack Webhook (auto-confirm payment)
  ├── Verify Paystack Deposit (manual fallback)
  ├── onOrderStatusChanged (email on status change)
  ├── onFirstOrderWelcome (welcome email)
  └── sendNewsletter (batch email blast)
```

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5. User-Facing Test Cases

> **Legend:** ✅ Pass · ❌ Fail · ⚠️ Partial

5.1 Navigation & Navbar

ID	Test Case	Steps	Expected Result
NAV-01	Logo navigates home	Click logo/PF text	Redirects to `/`
NAV-02	Nav links active state	Navigate to `/shop`	"Shop" link shows red underline, others do not
NAV-03	Nav underline hover	Hover over any nav link	Animated red underline slides in

ID	Test Case	Steps	Expected Result
NAV-04	Contact Us scrolls	Click "Contact Us" in nav	Smooth-scrolls to contact section on homepage; if on another page, navigates home first then scrolls
NAV-05	Mobile: PF text visible	View on mobile (≤375px)	"PF" text appears beside logo in brand red cursive font
NAV-06	Mobile: PerfectFooties text on desktop	View on desktop (≥900px)	Full "PerfectFooties" text shown
NAV-07	Hamburger menu opens	Click hamburger on mobile	Drawer slides in from right with all nav items
NAV-08	Drawer closes on link click	Click any item in drawer	Drawer closes and page navigates
NAV-09	Dark mode toggle	Click moon/sun icon	Theme switches; persists on reload
NAV-10	Cart icon shows count	Add item to cart	Cart icon badge shows quantity
NAV-11	Admin button visible to admin	Sign in as admin	Teal "Admin" pill button appears in desktop nav
NAV-12	Admin button hidden from regular users	Sign in as regular user	No admin button
NAV-13	User menu shows when signed in	Sign in with Google	Avatar/initials appear; click shows profile menu
NAV-14	Navbar becomes opaque on scroll	Scroll page down 50px+	Navbar background fills in with brand cream/dark colour
NAV-15	Navbar transparent at top	At page top	Navbar is transparent

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5.2 Homepage

ID	Test Case	Steps	Expected Result
HOME-01	Hero section renders	Load `/`	Full-width hero image with headline, sub-text, and CTA buttons
HOME-02	Shop Now CTA navigates	Click "Shop Now" hero button	Navigates to `/shop`
HOME-03	Collection cards display	Scroll to collections	Cards show image, collection name, short description

ID	Test Case	Steps	Expected Result
HOME-04	Card hover — red glow	Hover over a collection card	Warm red shadow appears; border turns red; "Explore" link turns red
HOME-05	Collection card navigates	Click a collection card	Navigates to the correct `/shop/:collectionId`
HOME-06	FAQ section renders	Scroll to FAQ	Questions listed; click expands answer; click again collapses
HOME-07	Contact section renders	Scroll to bottom	Name, email, message form; phone number; map or location info
HOME-08	Announcement banner	If admin has published active announcement	Banner visible at top of page
HOME-09	Newsletter subscribe (footer)	Enter email in footer subscribe field, submit	Success message shown; email saved to Firestore `subscribers`
HOME-10	Section underlines are red	View any section heading	Underline decoration is brand red, not cyan

5.3 Shop & Collections

ID	Test Case	Steps	Expected Result
SHOP-01	Shop page loads all collections	Navigate to `/shop`	All categories are displayed as browsable cards
SHOP-02	Collection page loads products	Click into a collection	Products listed with name, price, image
SHOP-03	Filter by type (if applicable)	Use any filter UI	Products narrow to matching type
SHOP-04	Out-of-stock products	View product with stock = 0	Item shows "Out of Stock" and cannot be added to cart
SHOP-05	Low stock indicator	Product with stock ≤ 5	Shows "Only X left" badge
SHOP-06	Wishlist from collection	Click heart on product card	Item added to wishlist; heart turns filled
SHOP-07	Niche collections visible	Navigate to niche/limited-edition section	Shows special collection items with countdown or status

5.4 Item Detail Page

ID	Test Case	Steps	Expected Result
ITEM-01	Product images display	Open item detail	Main image shown; thumbnail gallery if multiple
ITEM-02	Size selection	Click a size	Size is selected/highlighted
ITEM-03	Colour selection	Click a colour swatch	Colour selected; image updates if applicable
ITEM-04	Add to cart	Select size/colour, click "Add to Cart"	Item added; cart count increments; toast/snackbar confirms
ITEM-05	Add without size (footwear)	Click "Add to Cart" without selecting size	Error shown: "Please select a size"
ITEM-06	Made-to-order note	View leather product detail	Shows "Made to order — 10–14 days production + 2–5 days shipping" note in red-tinted box
ITEM-07	Wishlist from detail	Click heart icon on detail page	Added to wishlist
ITEM-08	Quantity selector	Increment/decrement	Quantity changes; cannot go below 1
ITEM-09	Back navigation	Click back	Returns to collection page

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5.5 Cart

ID	Test Case	Steps	Expected Result
CART-01	Items persist in cart	Add items, navigate away, return	Cart items still present
CART-02	Remove item	Click remove/trash icon	Item removed; total recalculates
CART-03	Update quantity	Change quantity in cart	Subtotal updates accordingly
CART-04	Empty cart state	Remove all items	"Your cart is empty" message with Shop CTA
CART-05	Cart total calculation	Add multiple items	Total = sum of (price × quantity)
CART-06	Navigate to checkout	Click "Proceed to Checkout"	Navigates to `/checkout`

ID	Test Case	Steps	Expected Result
CART-07	Cart count in navbar	Add 3 items	Badge shows "3"

5.6 Checkout & Payment

ID	Test Case	Steps	Expected Result
CHK-01	Checkout requires sign-in	Visit `/checkout` without auth	Prompt to sign in or guest checkout flow shown
CHK-02	Shipping form required fields	Submit without filling name/address	Validation errors shown per field
CHK-03	International shipping fields	Select non-Nigeria country	Province and postal code fields appear
CHK-04	Order summary visible	Load checkout page	Itemised list of cart items with prices shown
CHK-05	Shipping cost applied	Enter address	Shipping fee added to total
CHK-06	Production note on checkout	Leather/custom item in cart	Dispatch estimate shows "Production: 10–14 days + shipping: 2–5 days (local)"
CHK-07	Gift card redemption	Enter valid active gift card code	Discount applied; balance deducted shown
CHK-08	Invalid gift card rejected	Enter invalid/expired code	Error: "Gift card not found" or "Card already redeemed"
CHK-09	Loyalty points redemption	Sign in with user who has ≥50 pts; tick "Use points"	₦500 off per 50 pts deducted from total
CHK-11	Referral code from URL	Visit `/?ref=CODE`; go to checkout	Code pre-populated; ₦500 discount shown
CHK-12	Manual referral code entry	Type a referral code at checkout	Validates against Firestore; discount applied
CHK-13	Invalid referral code rejected	Enter random code	Error: "Referral code not found"
CHK-14	Paystack modal opens	Click "Pay Now"	Paystack payment modal appears with correct amount
CHK-15	Successful payment flow	Complete Paystack test payment	Order confirmed; redirected to `/thank-you`; email received in Mailtrap

ID	Test Case	Steps	Expected Result
CHK-16	Order saved to Firestore	After successful payment	Order appears in `users/{uid}/orders/` with status `confirmed`
CHK-17	Stock decremented	Complete purchase of item with stock	Stock count on product decremented by quantity
CHK-18	Payment failure handling	Use a declined test card	Error shown; order remains `pending`; user stays on checkout

5.7 Thank You Page

ID	Test Case	Steps	Expected Result
TY-01	Page loads after payment	Complete checkout	`/thank-you` loads with order summary
TY-02	Order reference number shown	View thank you page	Order ID (Firestore doc ID) displayed
TY-03	Red branding throughout	View page	All accent elements (checkmark, progress dots, CTA button) are brand red `#e3242b` — no cyan
TY-04	CheckCircle icon is red	View hero section	CheckCircle icon is red, not teal
TY-05	Progress tracker steps	View order progress section	Steps show: Placed → Confirmed → Production → Shipped → Delivered
TY-06	Production time text	View progress/info section	Shows "10–14 days production + 2–5 days shipping depending on location"
TY-07	WhatsApp card shown	View page	WhatsApp contact card with green border shown (NOT cyan)
TY-08	Continue Shopping button	Click "Continue Shopping"	Navigates to `/shop`
TY-09	Email receipt button	Click "Email Receipt"	Triggers receipt email to customer
TY-10	Download receipt	Click "Download Receipt"	PDF/printable receipt downloads
TY-11	Savings banner (if discount applied)	Checkout with discount	Gold savings banner shows total saved

5.8 Account & Order History

ID	Test Case	Steps	Expected Result
ACC-01	Account page requires auth	Visit `/account` without sign-in	Redirect to login or sign-in prompt
ACC-02	Profile tab shows user info	Sign in; visit `/account`	Display name, email, profile photo shown
ACC-03	Loyalty tier displayed	View profile	Shows tier: Fresh / Star Client / Master Patron + points balance
ACC-04	Referral code shown	View profile after first order	Unique referral code displayed with copy/share button
ACC-05	Orders tab shows orders	Click Orders tab	All past orders listed with status chip
ACC-06	Order status colour coding	View orders	Pending = orange, Confirmed = blue, Production = purple, Shipped = teal, Delivered = green
ACC-07	View Order Details	Click "View Details" on order card	Navigates to `/account/orders/:orderId` with full detail
ACC-08	Order detail page: items listed	Open order detail	Itemised products, prices, quantities shown
ACC-09	Order detail page: shipping address	Open order detail	Shipping address shown if applicable
ACC-10	Print shipping label	Click "Print Shipping Label"	Printable shipping label opens/downloads with brand styling
ACC-11	Download receipt from detail	Click receipt button	PDF receipt downloads with brand colours
ACC-12	Cancel order option	View pending order	"Request Cancellation" button visible; submits `cancellationRequests` doc
ACC-13	Wishlist tab shows saved items	Add items to wishlist; view Wishlist tab	All wished items listed with image and price
ACC-14	Add to cart from wishlist	Click "Add to Cart" in wishlist	Item added to cart
ACC-15	Remove from wishlist	Click remove in wishlist	Item removed; list updates
ACC-16	Review order after delivery	Status = `received`; click "Leave Review"	Review form appears; submit saves to `testimonials`

ID	Test Case	Steps	Expected Result
ACC-17	Review not available before delivery	Order status = `confirmed`	No review option shown for that order

5.9 Gift Cards

ID	Test Case	Steps	Expected Result
GC-01	Gift card info page	Navigate to `/gift-cards`	Page explains gift card programme
GC-02	Apply gift card at checkout	Enter valid active code	Balance applied as discount; remaining balance shown
GC-03	Partially used card	Use card partially	Balance reduced correctly; status becomes `partially_used`
GC-04	Fully redeemed card	Use remaining balance	Status becomes `fully_redeemed`; card rejected on future use
GC-05	Expired card rejected	Enter expired card code	Error: card expired
GC-06	Invalid code rejected	Enter random string	Error: card not found
GC-07	Only one card per order	Apply card; try adding second	Second card rejected

5.10 Custom Order

ID	Test Case	Steps	Expected Result
CO-01	Page accessible	Navigate to `/custom-order`	Form page loads with hero banner
CO-02	Contact fields pre-fill	Sign in before visiting	Name and email auto-filled from Google account
CO-03	Product type required	Submit without type	Error: "Please select a product type"
CO-04	Colour swatch selection	Click preset colour circle	Colour text and hex value update; swatch shows red border
CO-05	Custom colour picker	Click the colour picker input	Colour picker opens; selecting a colour updates hex and text
CO-06	Colour text field	Type "Forest Green"	Hex value attempts to auto-resolve to matching colour

ID	Test Case	Steps	Expected Result
CO-07	EU size field — footwear only	Select "Footwear" type	EU size dropdown appears
CO-08	EU size required for footwear	Select Footwear, submit without size	Error: "Please select your shoe size"
CO-09	EU size hidden for non-footwear	Select "Bag" type	EU size dropdown is NOT shown
CO-10	Notes character limit	Type 500+ characters	Input capped at 500; counter shows "500/500"
CO-11	Photo upload — click	Click drop zone	File picker opens
CO-12	Photo upload — drag & drop	Drag image file onto drop zone	Image accepted; thumbnail preview shown
CO-13	Photo preview thumbnails	Upload 2 photos	Two 80×80 thumbnails shown with remove button
CO-14	Remove photo	Click ✕ on thumbnail	Photo removed; slot freed
CO-15	Max 5 photos enforced	Upload 5 photos	Drop zone disappears; no more uploads accepted
CO-16	Non-image rejected	Drop a PDF or .docx	Not accepted
CO-17	WhatsApp opens on submit	Fill valid form; submit	Browser opens WhatsApp to PerfectFooties number with prefilled message
CO-18	WhatsApp message includes order ref	Signed-in user submits	Message includes "Order Ref: {orderId}"
CO-19	WhatsApp message includes photo URLs	Upload photos; submit	Photo URLs listed in WhatsApp message
CO-20	Order saved to Firestore	Signed-in user submits	Order appears in `users/{uid}/orders/` with `type: 'custom'`, `status: 'pending'`
CO-21	Order visible in admin queue	Submit custom order; check admin → Orders	Order appears with type "custom"
CO-22	Success state shown	Successful submission	Green checkmark page with order reference number

ID	Test Case	Steps	Expected Result
CO-23	Info box is red, not cyan	View form	Info note box has red tint background/border
CO-24	Loading state during submit	Click submit	Button shows spinner and "Saving & Opening WhatsApp..."
CO-25	Unauthenticated: no Firestore save	Submit without sign-in	WhatsApp still opens; no order in Firestore (no error shown)

5.11 Gallery

ID	Test Case	Steps	Expected Result
GAL-01	Gallery page loads	Navigate to `/gallery`	Gallery images display in grid
GAL-02	Image opens full view	Click a gallery image	Opens full-size preview or new tab
GAL-03	Gallery caption shows when present	View gallery item	Caption displays beneath image
GAL-04	Gallery page loads quickly	Navigate to `/gallery`	Page loads without long blank state

5.12 Blog

ID	Test Case	Steps	Expected Result
BLOG-01	Blog page loads	Navigate to `/blog`	Blog posts list appears
BLOG-02	Blog post opens	Click a blog card	Post details page opens
BLOG-03	Post image and content render	View blog post	Headline, image, body, and author/date show correctly
BLOG-04	Blog navigation	Use back or breadcrumb	Returns to `/blog` without losing state

5.13 Testimonials

ID	Test Case	Steps	Expected Result
TEST-01	Testimonials page loads	Navigate to `/testimonials`	All published testimonials displayed with star ratings

ID	Test Case	Steps	Expected Result
TEST-02	Submit review	Order status = `received`; go to account; click review	Review form shown
TEST-03	Star rating required	Submit review without rating	Error shown
TEST-04	Text required	Submit with only a rating	Error: review text required
TEST-05	Optional photo upload	Attach photo to review	Photo uploaded to Storage; shown in testimonial
TEST-06	Review submitted	Complete form; submit	Success message; testimonial saved to Firestore
TEST-07	Cannot review twice	Already reviewed an order	Review option no longer shown for that order

5.14 Our Story

ID	Test Case	Steps	Expected Result
STORY-01	Our Story page loads	Navigate to `/our-story`	Page content renders with brand history and imagery
STORY-02	Section headings show brand red underline	View page headings	Underlines are red, not cyan
STORY-03	Content readable on mobile	View page on mobile	Text and images scale correctly without overflow

5.15 Our Team

ID	Test Case	Steps	Expected Result
TEAM-01	Our Team page loads	Navigate to `/our-team`	Team cards and bios display correctly
TEAM-02	Team images render	Scroll page	Images load and are not broken
TEAM-03	Mobile layout works	View on mobile	Cards stack and text remains legible

5.16 Newsletter Subscribe

ID	Test Case	Steps	Expected Result
NS-01	Subscribe with valid email	Enter email in footer; submit	Success message: "You're subscribed!"
NS-02	Email saved to Firestore	Subscribe; check Firebase console	Doc in `subscribers` collection with email and `subscribedAt` timestamp

ID	Test Case	Steps	Expected Result
NS-03	Invalid email rejected	Enter "notanemail"; submit	Validation error shown
NS-04	Duplicate email	Subscribe with same email twice	Either silently accepted or "Already subscribed" message (no crash)
NS-05	Empty field rejected	Click subscribe with no email	Required field error

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5.17 Wishlist

ID	Test Case	Steps	Expected Result
WL-01	Add to wishlist	Click heart on product card	Heart fills; item saved
WL-02	Wishlist persists	Add item; close browser; reopen	Item still in wishlist (localStorage)
WL-03	Remove from wishlist	Click filled heart	Item removed; heart unfurls
WL-04	Wishlist icon in navbar	Have items in wishlist	Wishlist count badge visible
WL-05	Wishlist in account	Visit Account → Wishlist tab	All wishlisted items shown

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5.18 Loyalty Points & Referrals

ID	Test Case	Steps	Expected Result
LOY-01	Points earned after delivery	Order status set to `received`	`loyaltyPoints` increases by 15 on user doc
LOY-02	Points shown in account	View `/account`	Loyalty points balance and tier displayed
LOY-03	Redeem points at checkout	Have ≥50 pts; use at checkout	₦500 deducted per 50 pts; pts deducted from balance
LOY-04	Cannot redeem more than balance	Try to redeem 100 pts with only 50	Capped at available balance
LOY-05	Referral code generated	Complete first order	Referral code appears on account profile
LOY-06	Referral URL capture	Visit `/?ref=CODE`	Code stored in sessionStorage

ID	Test Case	Steps	Expected Result
LOY-07	Referral discount at checkout	Arrive via referral URL; checkout	₦500 discount applied; prompt shows referral code
LOY-08	Referrer gets points	Referred user completes order	Referrer's Firestore `loyaltyPoints` increases by 50
LOY-09	Referral code usage tracked	Apply referral code	`referralUses` document created; `totalUses` on code increments
LOY-10	Tier: Fresh (0 orders)	New user	Tier shown as "Fresh"
LOY-11	Tier: Star Client (2+ orders)	User with 2 completed orders	Tier shown as "Star Client"; 5% discount referenced
LOY-12	Tier: Master Patron (4+ orders)	User with 4+ completed orders	Tier shown as "Master Patron"; 10% discount + free engraving

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6. Admin Panel Test Cases

> Access: Sign in with an admin Google account, navigate to `/admin`.

6.1 Dashboard

ID	Test Case	Steps	Expected Result
DASH-01	Dashboard loads	Navigate to `/admin`	Dashboard section shown by default
DASH-02	Total orders stat	View stat cards	Shows correct count of all orders
DASH-03	Revenue stat	View stat cards	Shows sum of confirmed/received/completed order totals
DASH-04	Pending orders stat	View stat cards	Shows count of orders with `status: 'pending'`
DASH-05	Production queue stat	View stat cards	Shows count of orders with `status: 'production'` — NOT low stock count
DASH-06	Customer count	View stat cards	Shows count of documents in `users` collection
DASH-07	Daily orders chart	View charts	Bar chart shows orders per day for last 7 days
DASH-08	Revenue chart	View charts	Line/bar chart shows revenue trend

ID	Test Case	Steps	Expected Result
DASH-09	Recent orders list	View right panel	Last 3 orders shown with status chip
DASH-10	Quick action buttons	Click "Production Queue" button	Navigates to production tracker section
DASH-11	Low stock banner	Product with stock ≤ 2	Warning banner appears listing low-stock products

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6.2 Orders Management

ID	Test Case	Steps	Expected Result
ORD-01	All orders listed	Open Orders section	All orders from all users shown, newest first
ORD-02	Filter by status	Select a status filter	List narrows to matching status only
ORD-03	Filter by type	Select type filter	Shows only orders of that type (leather, retail, custom, etc.)
ORD-04	Search by name/email	Type customer name	Matching orders surface
ORD-05	Expand order row	Click chevron/expand icon	Expanded row shows: email, phone, items, notes, shipping address, photos (custom orders), discount details
ORD-06	Custom order: phone is WhatsApp link	Expand custom order row	Phone number is green underlined; clicking opens WhatsApp with prefilled specialist message
ORD-07	Custom order: email is mailto link	Expand custom order row	Email is red underlined; clicking opens email client with prefilled subject + body
ORD-08	Custom order: reference photos shown	Expand custom order row with photos	Photo thumbnails shown; clicking opens full image in new tab
ORD-09	Change order status	Use status dropdown on order row	Status updates in Firestore; status history logged
ORD-10	Status → confirmed triggers email	Change status to `confirmed`	Customer receives order confirmation email in Mailtrap
ORD-11	Status → production triggers email	Change status to `production`	Customer receives "In Production" email

ID	Test Case	Steps	Expected Result
ORD-12	Status → shipped triggers email	Change status to `shipped`	Customer receives "Shipped" email (with tracking link if set)
ORD-13	Status → received triggers email	Change status to `received`	Customer receives "Delivered" email
ORD-14	Add admin note	Click notes icon; type note	Note saved with timestamp; visible in expanded row
ORD-15	Send confirmation email manually	Click email icon on order	Confirmation email sent; success toast shown
ORD-16	Delete order	Click delete; confirm	Order removed from Firestore
ORD-17	Export to CSV	Click export button	CSV downloads with all visible orders
ORD-18	Create admin order	Click "Add Order"	Form appears; fill details; order saved to `users/admin-legacy/orders/`
ORD-19	Extra charge	Add extra charge to an order	Extra charge amount + reason saved; shown in expanded row
ORD-21	Tracking link	Add tracking link to shipped order	Tracking link saved; customer email includes "Track My Package" link

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6.3 Production Queue

ID	Test Case	Steps	Expected Result
PROD-01	Section loads	Navigate to Production Queue	Page shows spinner then loads orders
PROD-02	Shows confirmed orders	Order has status `confirmed`	Order card visible in queue
PROD-03	Shows production orders	Order has status `production`	Order card visible with "In Production" chip
PROD-04	Does NOT show pending/shipped orders	Orders with other statuses	NOT shown in production queue
PROD-05	Urgency: On Track	Order < 12 days old	Green "On Track" chip
PROD-06	Urgency: Due Soon	Order within 2 days of 14-day window	Orange "Due Soon" chip

ID	Test Case	Steps	Expected Result
PROD-07	Urgency: Overdue	Order older than 14 days	Red "Overdue" chip with warning icon
PROD-08	Progress bar	View order card	Blue/green/red bar shows % of 14-day production window used
PROD-09	Sort order	Multiple orders with different urgency	Overdue first, then Due Soon, then On Track
PROD-10	Mark Shipped	Click "Mark Shipped" button	Order status updated to `shipped`; card removed from queue; email sent to customer
PROD-11	Queue refreshes	After marking shipped	Queue reloads and shipped order is gone
PROD-12	Empty state	No confirmed/production orders	"Queue is clear!" message with green check icon

6.4 Customers

ID	Test Case	Steps	Expected Result
CUS-01	Customer list loads	Open Customers section	All users listed
CUS-02	Loyalty points shown	View customer row	Points balance, earned, redeemed visible
CUS-03	Tier shown	View customer row	Fresh / Star Client / Master Patron label
CUS-04	Order count shown	View customer row	Number of orders placed
CUS-05	Search customer	Type name or email	Matching customers shown

6.5 Collections & Products

ID	Test Case	Steps	Expected Result
COL-01	Collections load	Open Collections section	All categories listed with products
COL-02	Add product	Click "+ Add Product" in a category	Product form dialog opens
COL-03	Required fields validated	Submit empty product form	Validation errors shown

ID	Test Case	Steps	Expected Result
COL-04	Image upload	Upload product image	Image saved to Firebase Storage; URL saved to product
COL-05	Edit product	Click edit on existing product	Pre-filled form; save updates Firestore
COL-06	Delete product	Click delete; confirm	Product removed from category's products array
COL-07	Stock management	Set stock value	Stock saved; item shows "Out of Stock" at 0 on storefront
COL-08	Create category	Click "+ New Category"	Category added to Firestore
COL-09	Delete category	Delete with no products	Category removed

---

6.6 Newsletter Campaigns

ID	Test Case	Steps	Expected Result
NL-01	Newsletter section loads	Open Newsletter section	Two tabs: Subscribers and Campaigns
NL-02	Subscribers tab	Click Subscribers tab	Table of all subscriber emails with dates
NL-03	Search subscribers	Type email fragment	Filtered list shown
NL-04	Delete subscriber	Click delete on row	Subscriber removed from Firestore
NL-05	Copy all emails	Click "Copy All"	All visible emails copied to clipboard
NL-06	Compose new campaign	Click "+ New Campaign"	Dialog opens with split-panel (form left, preview right)
NL-07	Subject required	Submit without subject	Error: "Subject is required"
NL-08	Headline required	Submit without headline	Error: "Headline is required"
NL-09	Body required	Submit without body	Error: "Body text is required"
NL-10	Live preview updates	Type in any field	Right-panel preview updates in real time

ID	Test Case	Steps	Expected Result
NL-11	Preview: logo in header	View email preview	Dark header with PerfectFooties logo visible
NL-12	Preview: red accent bar	View email preview	Red bar below header
NL-13	Image upload — file	Switch to Upload tab; drag/drop image	Progress bar; image appears in preview
NL-14	Image upload — URL	Switch to URL tab; paste image URL	Image appears in preview
NL-15	CTA button optional	Leave CTA fields blank	Preview shows no button
NL-16	CTA button with text + URL	Fill CTA fields	Red button shown in preview
NL-17	Save as Draft	Click "Save Draft"	Campaign saved to Firestore with `status: 'draft'`; dialog closes
NL-18	Draft appears in Campaigns tab	After saving draft	Row in campaign table with "Draft" status chip
NL-19	Edit draft	Click edit icon on draft	Compose dialog reopens with existing values
NL-20	Send Now	Click "Save & Send"	Confirmation prompt; on confirm: emails sent to all subscribers
NL-21	Sent count shown	After send completes	Campaign row shows sent count and "Sent" status chip
NL-22	Email received	Check Mailtrap for each subscriber	Emails arrive with correct content, logo, red bar, CTA
NL-23	Cannot resend sent campaign	Click send on "Sent" campaign	Error or "Already sent" — button disabled
NL-24	Delete campaign	Click delete icon	Campaign removed from Firestore
NL-25	Image thumbnail in table	Campaign with image	Thumbnail shown in campaign table row

---

6.7 Gift Cards (Admin)

ID	Test Case	Steps	Expected Result
GCA-01	Gift cards list loads	Open Gift Cards section	All gift cards shown with status and balance

ID	Test Case	Steps	Expected Result
GCA-02	Create gift card	Click "+ Create Card"	Card generated with 8-character code; status `pending`
GCA-03	Activate card	Click "Activate" on pending card	Status → `active`; expiry date set 1 year from now
GCA-04	View transactions	Expand gift card row	Shows list of redemption transactions with amounts and order IDs
GCA-05	Export gift card data	Click export	CSV/data export downloads

6.8 Loyalty & Referrals (Admin)

ID	Test Case	Steps	Expected Result
LAR-01	Referral leaderboard	Open Loyalty section	List of referral codes ranked by total uses
LAR-02	Manual point adjustment	Select user; enter delta and reason	`loyaltyAdjustments` doc created; user's `loyaltyPoints` updated
LAR-03	Negative adjustment	Enter a negative delta	Points reduced; cannot go below 0
LAR-04	Adjustment logged	After adjustment	Adjustment visible in adjustment history with before/after balance
LAR-05	All users loyalty list	View all users	Points, tier, earned, redeemed shown per user

6.9 Blog Posts

ID	Test Case	Steps	Expected Result
BLOG-01	Blog post list	Open Blog Posts section	All posts listed
BLOG-02	Create post	Click "+ New Post"; fill all fields	Post saved to Firestore with auto-generated slug
BLOG-03	Edit post	Click edit	Pre-filled form; save updates doc
BLOG-04	Delete post	Click delete; confirm	Post removed
BLOG-05	Post appears on /blog	Publish post	Visible at `/blog` with correct content

6.10 Gallery

ID	Test Case	Steps	Expected Result
GAL-01	Gallery images list	Open Gallery section	All gallery images shown
GAL-02	Upload image	Click upload; select file	Image saved to Storage; URL in Firestore
GAL-03	Add caption	Fill caption field	Caption saved and displayed on gallery page
GAL-04	Delete image	Click delete; confirm	Image removed from Firestore (and ideally Storage)
GAL-05	Reorder images	Drag or set order number	Order field updates; gallery page reflects new order

---

6.11 Announcements

ID	Test Case	Steps	Expected Result
ANN-01	Create announcement	Fill title, message, CTA	Announcement saved
ANN-02	Activate announcement	Toggle active = true	Banner appears on homepage
ANN-03	Deactivate announcement	Toggle active = false	Banner removed from homepage
ANN-04	Expiry date	Set expiry in past	Announcement no longer shown even if active
ANN-05	CTA link	Set ctaUrl	Banner has clickable CTA button

---

7. Cloud Function & Email Test Cases

> Inspect emails at: Mailtrap inbox.

ID	Test Case	Trigger	Expected Email
EML-01	Confirmation email via webhook	Complete Paystack payment	Subject: "Order Confirmed"; includes items, total, order ID; PerfectFooties logo in header
EML-02	Confirmation email via admin	Admin clicks email icon on order	Same confirmation template sent manually

ID	Test Case	Trigger	Expected Email
EML-03	Production email	Admin sets status → `production`	Subject: "Your Order is Being Crafted"; includes order ID and item name
EML-04	Shipped email	Admin sets status → `shipped` or `shipping`	Subject: "Your Order is On Its Way"; includes tracking link if present
EML-05	Shipped email with tracking	Admin adds tracking link before shipping	Email shows "Track My Package" button
EML-06	Delivered email	Admin sets status → `received` or `delivered`	Subject: "Your Order Has Arrived!"; full itemised receipt included
EML-07	Welcome email	First ever order confirmed	One-time welcome email sent (does NOT repeat on subsequent orders)
EML-08	Newsletter batch send	Admin sends newsletter from campaign	All subscribers receive branded email with correct headline, body, image, CTA
EML-09	Logo in all emails	Open any email	PerfectFooties logo (72px circular) visible in dark gradient header
EML-10	Production timeframe in emails	Open confirmation email	Shows "10–14 days production + 2–5 days shipping" — NOT "5–10 business days"
EML-11	No email without address	Order with no email field	Email function exits gracefully; no crash in logs
EML-12	Webhook signature validation	Send request to webhook URL without valid signature	Returns 401; no order update

---

8. Payment Flow Test Cases

> Use Paystack test cards. Test mode dashboard: https://dashboard.paystack.com/#/test

Paystack Test Cards

Card Number	CVV	Expiry	Result
4084 0840 8408 4081	408	Any future	Success
4084 0840 8408 4081	408	Any future	Success (PIN: 0000)
5531 8866 5214 2950	564	Any future	Success
4084 0840 8408 4081	Any	Any past	Declined

ID	Test Case	Steps	Expected Result
PAY-01	Full payment for retail order	Add retail product; checkout; pay	Paystack modal shows full total; on success → order confirmed

ID	Test Case	Steps	Expected Result
PAY-02	Webhook confirms order	Pay via test card	Within ~30 seconds: order in Firestore updates to `confirmed`, `depositVerified: true`
PAY-04	Manual verify fallback	Webhook missed; click "Verify Payment"	`verifyPaystackDeposit` callable checks Paystack API; updates order if paid
PAY-05	Amount mismatch rejected	Tampered amount in request	`verifyPaystackDeposit` rejects if paid amount < expected
PAY-06	Duplicate webhook ignored	Webhook fires twice for same reference	Second call sees `depositVerified: true` and skips; no double email
PAY-07	Declined payment	Use declined test card	Error message shown in Paystack modal; order stays `pending`
PAY-08	Order with discount paid	Apply gift card / loyalty / referral; pay	Discounted total charged via Paystack; Firestore records discount amounts

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9. Security & Access Control Tests

ID	Test Case	Expected Result
SEC-01	Regular user cannot access `/admin`	Sign in as regular user; visit `/admin`
SEC-02	Unauthenticated cannot read orders	Direct Firestore SDK call without auth
SEC-03	User cannot read another user's orders	User A tries to read User B's orders path
SEC-04	User can read own orders	Authenticated user reads own orders
SEC-05	Admin can read all orders via collection group	Admin queries collection group
SEC-06	Non-admin cannot send newsletter	Regular user calls `sendNewsletter` callable
SEC-07	Unauthenticated newsletter send blocked	Call `sendNewsletter` without auth
SEC-08	Newsletter collection: admin read only	Regular user reads `newsletters`
SEC-09	Subscribers: anyone can create	POST subscriber without auth

ID	Test Case	Expected Result
SEC-10	Subscribers: only admin can delete	Regular user tries to delete subscriber
SEC-11	Storage: custom-orders only auth users	Unauthenticated upload to custom-orders/
SEC-12	Storage: product images admin only	Regular user tries to upload to products/
SEC-13	Gift cards: admin write only	Regular user tries to create gift card
SEC-14	Order status validation	Try to set order status to `invalid-status`
SEC-15	Paystack webhook signature	POST to webhook without x-paystack-signature

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10. Responsive / Mobile Tests

> Test on actual devices or Chrome DevTools at the sizes below.

Breakpoint	Width	Test Focus
Mobile S	320px	No horizontal overflow; text readable
Mobile M	375px	Standard iPhone; all interactions reachable
Mobile L	425px	Larger phones
Tablet	768px	Two-column layouts where applicable
Desktop	1440px	Full navigation, side panels

ID	Test Case	Expected Result
RES-01	Navbar on mobile	Hamburger menu visible; no nav links shown; PF text beside logo
RES-02	Mobile drawer	All nav items accessible in drawer; close button works
RES-03	Cart page on mobile	Items stack vertically; remove button accessible
RES-04	Checkout form on mobile	All inputs reachable; no clipping
RES-05	Admin panel on mobile	Sidebar collapses; content scrollable
RES-06	Newsletter compose on mobile	Edit/Preview tabs shown instead of split panel
RES-07	Custom order form on mobile	Photo grid wraps; drop zone visible
RES-08	Thank you page on mobile	Progress tracker readable; buttons full width

ID	Test Case	Expected Result
RES-09	Product detail on mobile	Image gallery scrollable; add to cart button fixed or visible
RES-10	Bottom mobile nav	Fixed bottom nav bar visible on mobile with key icons

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11. Edge Cases & Negative Tests

ID	Scenario	Expected Behaviour
EDGE-01	Order with 0 items	Should not be possible; checkout disabled if cart empty
EDGE-02	User has no email address	Emails skipped gracefully; function logs warning but no crash
EDGE-03	Network offline during payment	Paystack handles; order stays pending; user can retry
EDGE-04	Admin deletes order mid-production	Order disappears from queue on next load
EDGE-05	Two users place last-in-stock item simultaneously	First Paystack confirmation wins; second gets error or partial fulfilment
EDGE-06	Gift card with £0 balance	Rejected: "Card fully redeemed"
EDGE-07	Loyalty redemption exceeds order total	Cannot redeem more than the order value
EDGE-08	Referral code used by code owner	Should either be rejected or simply not apply (self-referral blocked)
EDGE-09	Newsletter to 0 subscribers	Function returns `{ success: true, sentCount: 0 }` gracefully
EDGE-10	Custom order without sign-in	No Firestore save; WhatsApp still opens; no crash
EDGE-11	Upload non-image to custom order	File rejected; error shown
EDGE-12	Upload 6th photo	Slot capped at 5; no more uploads accepted
EDGE-13	Very long customer name (200+ chars)	`sanitizeString` trims at 200 chars
EDGE-14	SQL-like characters in fields	Sanitized; not injectable (Firestore is NoSQL but XSS possible)

ID	Scenario	Expected Behaviour
EDGE-15	Open WhatsApp link on desktop	Opens WhatsApp Web
EDGE-16	Order placed then cancelled by user	`cancellationRequests` doc created; order NOT auto-deleted
EDGE-17	Admin marks production order as shipped without tracking link	Email sent without tracking button; no crash
EDGE-18	Paystack webhook fires before order saved	Rare race condition; webhook finds no order; returns 200 OK silently
EDGE-19	Dark mode + admin panel	Admin panel respects theme; all text readable
EDGE-20	Rapid double-click on "Add to Cart"	Item not added twice (debounce or one-at-a-time write)

12. Known Constraints & Out of Scope

Item	Notes
International shipping rates	Currently not dynamically calculated; flat estimates shown
Apple Pay / Google Pay	Not implemented; Paystack card only
Order search by reference ID	Not available on customer-facing account page
Push notifications	Notification bell exists; deep integration may vary
SMS notifications	Not implemented
Inventory sync across sessions	Stock decremented on Paystack confirmation; concurrent stock race is a known risk
Blog comments	Not implemented
Multi-currency	NGN only
Pagination	Large order lists loaded without pagination (may slow at scale)
Staging environment	No dedicated staging; all tests hit production Firebase project — use test Paystack mode and test user accounts

13. Test Data Requirements

Paystack

- Use **test mode** API keys (confirm with dev team)

- Test cards listed in Section 8

Firebase Test Data to Prepare Before Testing

Data	How to Create	Used For
At least 3 products with stock	Admin → Collections	Cart, checkout, stock decrement tests
1 product with stock = 0	Admin → Collections	Out-of-stock display
1 active gift card	Admin → Gift Cards → Create + Activate	Gift card redemption
1 expired gift card	Admin → Gift Cards → Create with past expiry	Expired card rejection
2 regular Google test accounts	Google accounts	User A & User B for referral test
1+ newsletter subscribers	Subscribe via footer	Newsletter send test
1 draft newsletter campaign	Admin → Newsletter → New Campaign → Save Draft	Send newsletter test
1 custom order	Place via `/custom-order` with test account	Admin custom order view

Environment Variables (confirm with dev)

```
VITE_FIREBASE_API_KEY
VITE_FIREBASE_PROJECT_ID
VITE_FIREBASE_STORAGE_BUCKET
VITE_FIREBASE_APP_ID
VITE_PAYSTACK_PUBLIC_KEY   ← must be TEST key for QA
```

---

14. Bug Reporting Template

Use the following format for every defect found:

```
BUG-###

Title: [Short, clear description]

Environment:
  - URL:
  - Browser & Version:
  - Device / Screen Size:
  - User Role: (Guest / Regular User / Admin)
```

```
Test Case ID: [e.g. CHK-15]

Steps to Reproduce:
  1.
  2.
  3.

Expected Result:
  [What should happen according to this document]

Actual Result:
  [What actually happened]

Severity: Critical / High / Medium / Low
  - Critical = blocks core user flow (payment, order placement)
  - High = major feature broken but workaround exists
  - Medium = incorrect behaviour but minor impact
  - Low = cosmetic or UI issue

Screenshots / Recording:
  [Attach]

Firebase / Console Errors (if any):
  [Paste any Firestore errors, function logs, or browser console errors]

Notes:
  [Any additional context]
```

Appendix A — Status Lifecycle

```
Retail / Leather / Custom Orders:
  pending → confirmed → production → shipped/shipping → received/delivered
                                                      ↘ cancelled

Gift Cards:
  pending → active → partially_used → fully_redeemed
                                    → expired (by date)
```

Appendix B — Brand Colour Reference

Token	Hex	Used For
Brand Red	#e3242b	CTAs, active states, accents, icons
Brand Red Dark	#b81b21	Hover states on red elements
Teal	#007a7a	Admin button, secondary actions

Token	Hex	Used For
Warm Beige	`#E8D5B0`	Card borders, dividers
Background Cream	`#FFF8F0`	Page background
WhatsApp Green	`#25D366`	WhatsApp buttons only

> Any element that appears **cyan/teal** on a customer-facing page (outside of admin teal buttons) is a bug. All customer-facing accents must be brand red.

End of QA Test Plan — PerfectFooties v1.0